



Dominant Psychosocial Response and Distress Classification in Oncology-Related Messages

A Synthetic, Label-Leakage-Controlled NLP Study

K.T.

Holon Institute of Technology (HIT) — Faculty of Digital Medical Technologies

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Abstract

We study two-label classification of oncology-related messages: the dominant *psychosocial response* (seven classes: anxiety, sadness, anger, hope, guilt, denial, acceptance) and the *intensity of distress* (three ordinal levels: low, medium, high). We found no public corpus with this exact joint seven-response and three-level distress framing, and emotional labels are subjective and costly to annotate; we therefore construct a fully synthetic corpus using attribute-based generation under a strict *label-leakage ban* (the generated text never explicitly names the target response label), and we audit every item with two independent large-language-model judges, organizing the corpus into strict, silver, and consensus quality tiers. We generated 2,033 candidate messages, obtained 2,025 valid audited messages, and retained 1,273 model-ready items across these tiers. We compare three model families on both tasks — a sparse lexical baseline (TF-IDF with logistic regression), a fine-tuned transformer (DistilBERT), and an off-the-shelf zero-shot model (BART-MNLI) — using macro-F1, ordinal-aware metrics, and a 5000-resample paired bootstrap, and we validate the synthetic labels against a blind human annotator on a 67-item test set. On response classification a lexical model has the highest point estimate (macro-F1 0.856), while its difference from the best transformer is not statistically resolved; on distress the transformer leads on the mean (0.864) but its advantage over the lexical model is not statistically resolved on 67 items, whereas both supervised systems reliably outperform zero-shot. We further find that more synthetic data is not uniformly better, and that residual errors concentrate at the low/medium distress boundary rather than at severe failures.

1 Introduction

A cancer diagnosis carries a heavy psychosocial burden for patients and caregivers, and supportive-care teams increasingly seek to triage the emotional state behind patient-generated text such as forum posts, messages, and intake notes. Two questions are clinically relevant: *which psychosocial response dominates* a message, and *how intense the distress* is. Automating this at scale is attractive, but two obstacles stand in the way. First, there is no public labeled dataset for this exact joint framing. Second, emotional labels are subjective and expensive to obtain, and a naive classifier can “cheat” on explicit emotion keywords rather than learning the underlying state.

This project addresses both obstacles by generating its own data and by studying which modelling approach is actually justified for the task. Our contributions are:

- a novel two-label task on oncology messages (seven-class psychosocial response and a three-level ordinal distress label), distinct from broad sentiment classification;
- a fully synthetic, attribute-controlled corpus built under a strict label-leakage ban and audited by two independent LLM judges into strict/silver/consensus quality tiers;

Dominant Psychosocial Response & Distress Classification in Oncology Messages

A fully synthetic, label-leakage-controlled NLP benchmark: 7 psychosocial responses + 3 distress levels

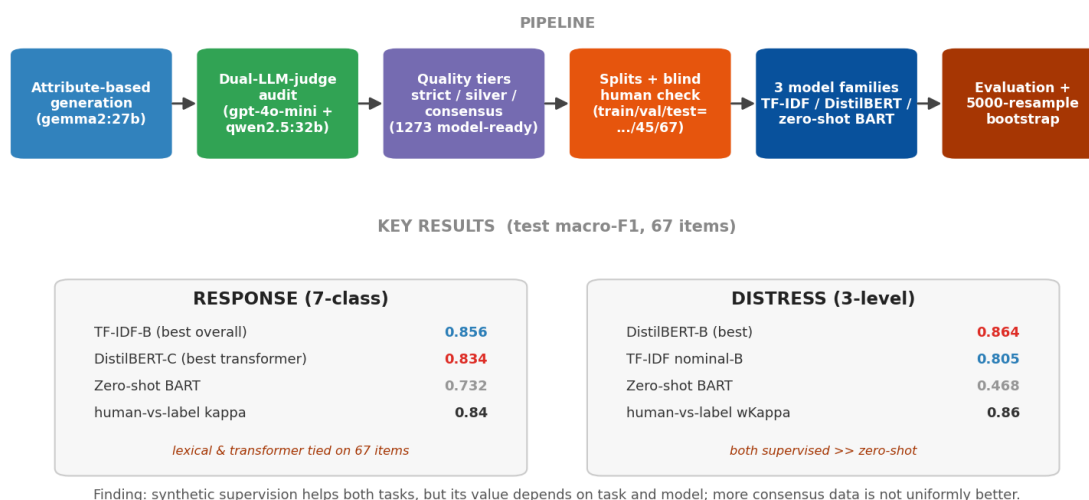


Figure 1: Overview of the pipeline (attribute-based generation, dual-LLM-judge audit, quality tiers, splits with a blind human check, three model families, and bootstrap evaluation) and the headline test results for both tasks.

- a controlled comparison of three model families on both tasks, with significance testing via a 5000-resample paired bootstrap and an independent blind human validation; and
- an analysis of how the amount of synthetic data affects each model, plus a consolidated error analysis identifying where the remaining errors lie.

Figure 1 summarizes the pipeline and the headline results.

2 Related Work

Our method draws on three lines of recent work and is most directly comparable to one domain study. Yu et al. (2023) introduce attribute-conditioned synthetic data generation, in which a language model is prompted with sampled attributes to control diversity; they show that attribute diversity improves downstream utility but that generation can introduce systematic bias — exactly the trade-off we manage when sampling oncology attributes. Li et al. (2023) study when LLM-generated synthetic data helps text classification and report that synthetic data becomes *less effective as task subjectivity rises*; because emotional labels are highly subjective, this motivates our strict auditing, quality tiers, and human check. Zheng et al. (2023) formalize the LLM-as-judge paradigm, showing that strong LLM judges reach over 80% agreement with human experts while exhibiting position, verbosity, and self-enhancement biases; this provides a methodological precedent for scalable LLM-based evaluation and documents the biases that motivate our use of two independent judges and a human validation. Finally, Xu et al. (2026) classify emotional tone in online cancer peer-support posts using LLM-based annotation and a range of models (TF-IDF logistic regression, random forest, LightGBM, GRU, and a fine-tuned ALBERT) over three tone classes; this is the closest domain baseline. Our task differs in that we predict seven fine-grained psychosocial responses (rather than negative/neutral/positive), add a separate ordinal distress task, use a fully synthetic attribute-controlled corpus with dual-judge auditing and quality tiers, and explicitly analyze human ambiguity and the divergence between the synthetic annotation policy and human interpretation. A broad survey of NLP for mental-illness detection (Zhang et al., 2022) situates the wider domain.

Finally, this project continues a line of work at the host institution on constructing controlled synthetic

benchmarks for evaluating language models on text classification. Aperstein et al. (2025) build a synthetic benchmark through controlled paraphrasing and deliberate information removal and compare discriminative and generative models for classifying semantic relations between text pairs, releasing both the dataset and its generation code; our attribute-controlled corpus and three-family model comparison apply the same methodology to psychosocial response and distress. In the clinical domain, Werthaim et al. (2026) generate synthetic patient cases and simulate interactions with GPT-4.1 and GPT-4o-mini agents to benchmark diagnostic questioning, and Mama et al. (2025) benchmark language models on noisy patient narratives; these motivate, respectively, our use of an LLM judge (gpt-4o-mini) over synthetic medical text and our explicit modelling of message noise.

3 Task and Data

3.1 Task definition

Given a single English oncology-related message, we predict two labels with separate models: $f(\text{text}) \rightarrow (\text{response} \in \mathcal{R}, \text{distress} \in \{\text{low}, \text{medium}, \text{high}\})$, where $\mathcal{R} = \{\text{anxiety, sadness, anger, hope, guilt, denial, acceptance}\}$. Distress is ordinal: the error low→high is treated as worse than low→medium, so distress is evaluated with ordinal-aware metrics.

3.2 Attribute-based synthetic generation

The corpus is fully synthetic (no real patient data). Each message is generated from a sampled combination of attributes — role (patient/caregiver/...), cancer stage, cancer type, tone, channel, age group, length, and a noise flag — so that the attribute grid enforces diversity and coverage. A local generator (gemma2:27b via Ollama) is conditioned on the sampled attributes *and* an intended (response, distress) label and asked to write a natural message that expresses that state *without ever explicitly naming the target response label*. This label-leakage ban reduces the risk that a classifier succeeds by detecting explicit class-name tokens. Malformed outputs and explicit constraint violations were rejected. Label drift was audited separately: items were either rejected or retained with consensus labels when both judges confidently agreed on an alternative interpretation.

3.3 Dual-LLM-judge auditing and quality tiers

Two independent judges (gpt-4o-mini and qwen2.5:32b) read each message *without* seeing the intended label and predict the two labels. Each item is assigned to a quality tier. Strict items matched both intended labels according to both judges with confidence at least 4 and no ambiguity. Silver items also matched both intended labels according to both judges, but had confidence at least 3 rather than meeting the strict threshold. Consensus items were those for which both judges confidently agreed with each other on labels that differed from the intended labels; their final labels were set to the judge consensus. The union of these three tiers forms the model-ready corpus (Table 1).

3.4 Splits and human validation set

We build a frozen test set of 67 strict-tier items and a validation set of 45 items (seed 42), with three nested training tiers: *train_A* (strict only, 359), *train_B* (strict+silver, 434), and *train_C* (strict+silver+consensus, 1161). The three tiers let us measure whether more synthetic data of decreasing purity helps. The 67-item test set was independently re-annotated by a human (Section 5.6) to validate the synthetic labels before any model was trusted.

Table 1: Corpus construction funnel and the data splits (seed 42). Only the model-ready subset was used in the supervised experiments.

Corpus construction		Data splits	
Stage	Count	Split	Size
candidate messages	2033	test	67
valid audited	2025	validation	45
model-ready (used)	1273	train_A (strict)	359
strict	471	train_B (+silver)	434
silver	75	train_C (+consensus)	1161
consensus	727		
review-only	752		
needs-rejudging	8		

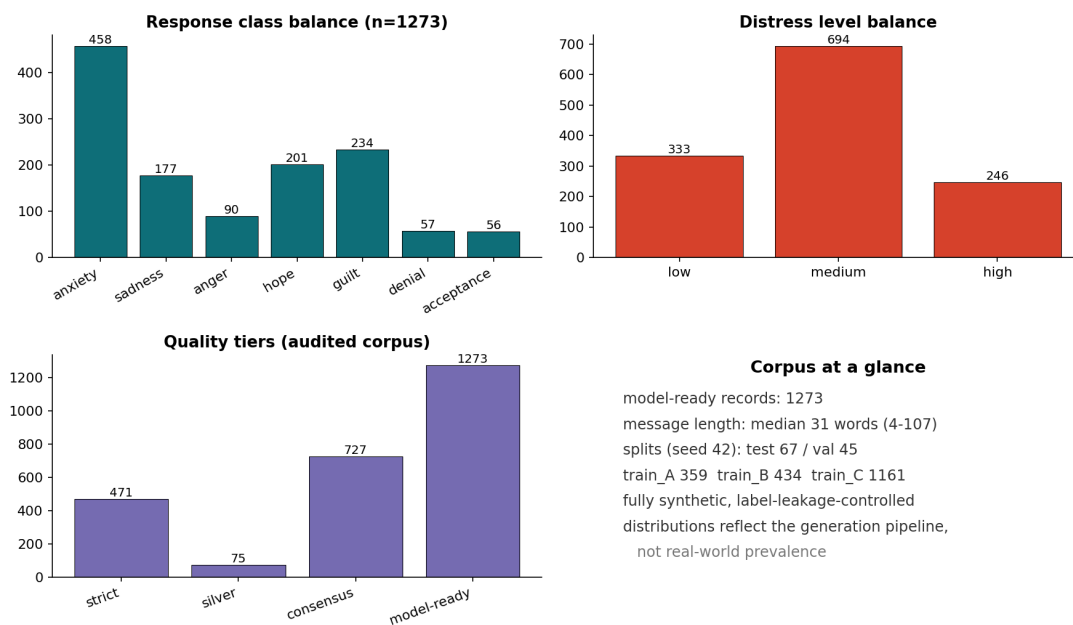


Figure 2: Exploratory data analysis of the model-ready corpus: response and distress class balance, quality-tier sizes, and summary statistics.

3.5 Exploratory data analysis

The model-ready corpus ($n = 1273$) is class-imbalanced. For response, the class counts are anxiety 458, guilt 234, hope 201, sadness 177, anger 90, denial 57, and acceptance 56; for distress they are low 333, medium 694, and high 246. Messages are short (median 31 words; range 4–107). Figure 2 summarizes the distributions. We stress that these distributions reflect the generation and auditing pipeline and are *not* estimates of real-world prevalence.

4 Methods

4.1 Models

We compare three model families on both tasks, alongside a trivial floor:

- **Majority class:** predicts the most frequent label (sanity floor).
- **TF-IDF + logistic regression:** class-weighted multinomial logistic regression on TF-IDF features.

For distress we evaluate two variants: a *nominal* three-class model and an *ordinal* two-threshold formulation ($P(y > \text{low})$, $P(y > \text{medium})$) with validation-tuned thresholds and a monotonic correction.

- **DistilBERT** (distilbert-base-uncased): a fine-tuned transformer with a classification head, trained separately for each task with class-weighted cross-entropy.
- **Zero-shot BART-MNLI** (facebook/bart-large-mnli): off-the-shelf natural-language-inference zero-shot classification (two prompt templates averaged for distress).

4.2 Training and hyperparameters

DistilBERT is fine-tuned with the regime in Table 2, identical for both tasks. A separate model is trained for each tier (A/B/C), and all transformer results are reported as the mean and standard deviation over five random seeds.

Table 2: DistilBERT fine-tuning configuration (identical for both tasks).

Setting	Value
Base model	distilbert-base-uncased
Seeds	13, 42, 73, 101, 2026 (mean $\pm$ std)
Max epochs	12, early stop (patience 2, $\Delta = 0.005$ val macro-F1)
Batch size	16
Learning rate	2×10^{-5} (linear schedule, warmup 0.10)
Weight decay / grad clip	0.01 / 1.0
Max token length	128
Loss	class-weighted cross-entropy (weight cap 5.0)
Model selection	best validation macro-F1

4.3 Evaluation metrics

The primary metric is **macro-F1**, which gives equal weight to every class and is therefore appropriate for evaluating imbalanced multiclass data. For response we also report micro/macro precision, recall, and F1, and a minority macro-F1 over the rare classes (anger, denial, acceptance). For distress, being ordinal, we report linear and quadratic weighted Cohen’s κ , mean absolute ordinal error (MAE), exact accuracy, the *severe-error rate* (low $\leftrightarrow$ high), the adjacent-error rate, and per-level scores. Statistical significance is assessed with a **5000-resample stratified paired bootstrap**: we report per-system 95% confidence intervals (CIs) and paired difference CIs, and call a difference reliable only if its CI excludes zero. For bootstrap comparisons involving DistilBERT, predictions from the five seeds were combined by majority vote; the bootstrap point estimate may therefore differ slightly from the mean of the five per-seed macro-F1 scores. Finally, we measure human agreement with Cohen’s κ (response) and weighted κ (distress) between a blind human annotator and the synthetic labels on the 67-item test set.

5 Results

5.1 Response classification

Figure 3 reports macro-F1 for all response systems. The strongest system is a sparse lexical model, TF-IDF+logistic regression trained on tier B (macro-F1 **0.856**), narrowly above the best transformer, DistilBERT on tier C (0.834). Zero-shot BART reaches 0.732 and the majority floor is 0.040. As Section 5.5 shows, the lexical and transformer systems are not statistically separable on the 67-item test set.

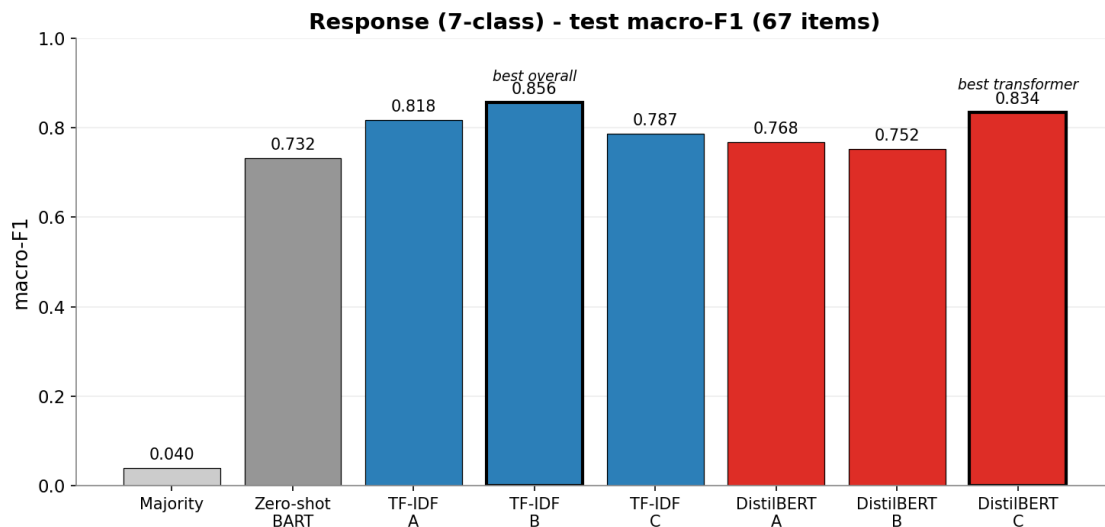


Figure 3: Response classification: test macro-F1 for all systems (67 items). Bold outlines mark the best lexical system (TF-IDF-B) and the best transformer (DistilBERT-C).

5.2 Distress classification

Figure 4 reports macro-F1 for all distress systems. DistilBERT on tier B achieves the highest mean (macro-F1 **0.864**, linear weighted κ 0.851, MAE 0.113). The best lexical model is TF-IDF nominal on tier B (0.805); the ordinal TF-IDF variant is consistently *weaker* than the nominal one (e.g. 0.786 vs. 0.805 on tier B), an honest negative result that may reflect threshold tuning on only 45 validation items or limitations of the two-threshold decomposition. Zero-shot BART substantially underperforms at 0.468 and is the only system with non-trivial severe errors; the majority floor is 0.211.

5.3 Effect of the data tier

Figure 5 contrasts macro-F1 across the three tiers for both model families. The peak tier depends on both the task and the model: for response, DistilBERT achieved its highest mean on tier C (0.768/0.752/0.834 for A/B/C), although the C-versus-A difference was not statistically resolved, while TF-IDF peaks at B (0.818/0.856/0.787); for distress, both families peak at B (DistilBERT 0.806/0.864/0.827; TF-IDF nominal 0.793/0.805/0.802), and the additional consensus data in C slightly hurts. In short, *more synthetic data is not uniformly better*: the lower-purity consensus tier was associated with a higher mean in one configuration, but transferred annotation-policy artifacts or imbalance in others.

5.4 Class-weighting control

To isolate the contribution of class weighting on the best distress configuration, we retrain DistilBERT-B with and without class weights under identical splits, seeds, and hyperparameters (Table 3). Weighting yields a modest but consistent gain (macro-F1 +0.023, weighted κ +0.023, MAE −0.015), concentrated on the harder low level, with no change to the zero-severe-error profile.

5.5 Statistical significance

Figure 6 shows per-system 95% bootstrap CIs for distress. DistilBERT-B (0.853, [0.758, 0.941]) and TF-IDF nominal-B (0.805, [0.709, 0.899]) overlap substantially, and their paired difference CI includes zero ([−0.066, +0.164]): the transformer’s mean advantage is *not* resolved on 67 items. Both supervised

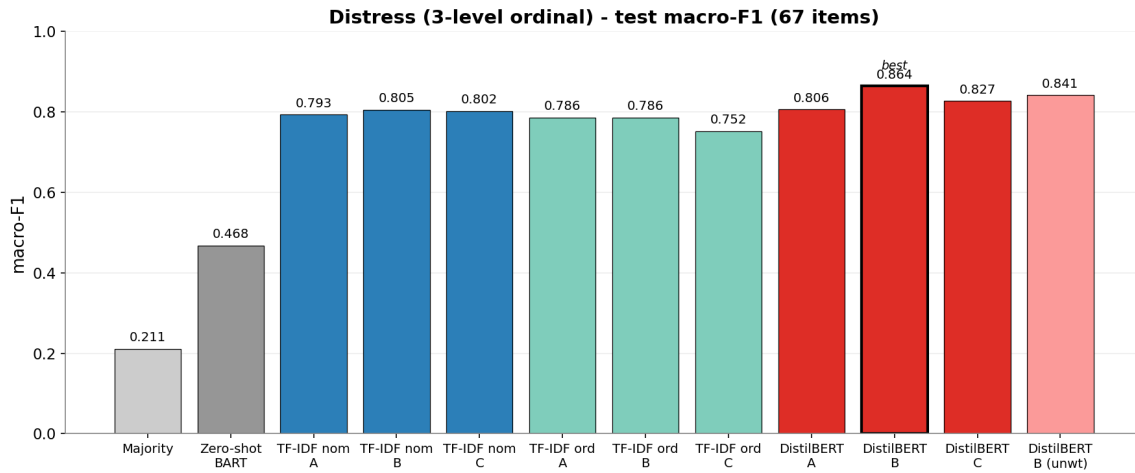


Figure 4: Distress classification: test macro-F1 for all systems (67 items), including nominal and ordinal TF-IDF variants and an unweighted DistilBERT-B control. The bold outline marks the best system (DistilBERT-B).

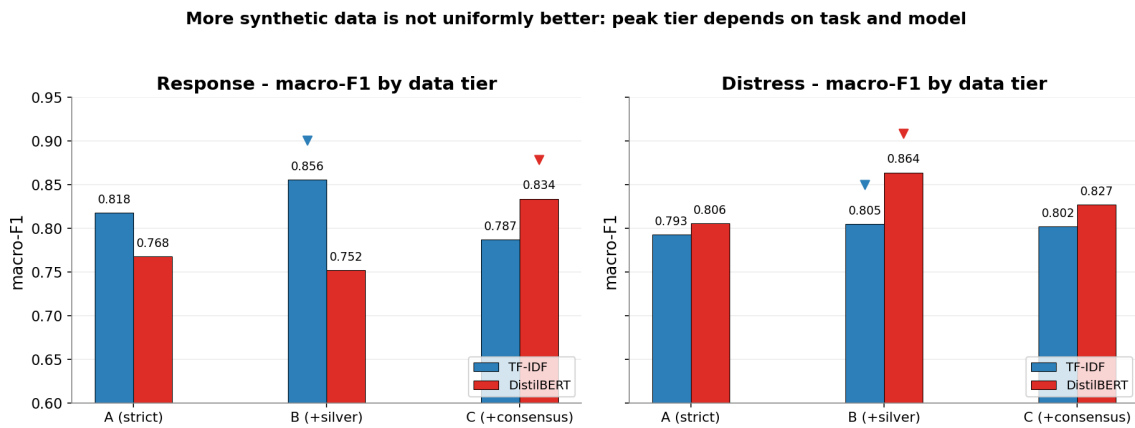


Figure 5: Macro-F1 by data tier (A: strict, B: +silver, C: +consensus) for both model families. The peak tier (marked) differs across tasks and models.

systems, however, separate cleanly from zero-shot (0.464, [0.336, 0.592]); for example, DistilBERT-B vs. zero-shot is +0.391 ([+0.228, +0.555]). For response, the only reliably separable pair is TF-IDF-B vs. zero-shot (+0.123, [+0.008, +0.241]); lexical and transformer systems are not separable. Supervised systems achieved higher point estimates than zero-shot in both tasks. Bootstrap-confirmed superiority was observed for TF-IDF-B in response and for both TF-IDF-B and DistilBERT-B in distress. The lexical-versus-transformer comparison remained underpowered at this test-set size.

5.6 Human validation

A single blind annotator re-labeled the 67-item test set. Agreement with the synthetic labels was high: for response, exact agreement was 86.6% (Cohen's $\kappa = 0.84$); for distress, exact agreement was 83.6% (weighted $\kappa = 0.80$ linear, 0.86 quadratic). Comparing the best model to the human reference on distress gives macro-F1 0.707 and weighted κ 0.661. Crucially, agreement was strong on human-unambiguous items and much weaker on human-ambiguous items (weighted κ 0.817 vs. 0.365), with the disagreement dominated by a systematic human-low $\rightarrow$ model-medium shift (Figure 7). High distress is recognized very reliably, and there are essentially no severe (low $\leftrightarrow$ high) errors. The distinction between low and medium distress is thus the least reliable part of the label scheme and remained sensitive to subjective interpretation.

Table 3: Class-weighting control on distress DistilBERT-B (identical split, seeds, hyperparameters).

Metric	Weighted-B	Unweighted-B	Δ
macro-F1	0.864	0.841	+0.023
weighted κ (linear)	0.851	0.828	+0.023
MAE (ordinal)	0.113	0.128	-0.015
severe (low $\leftrightarrow$ high)	0.000	0.000	—

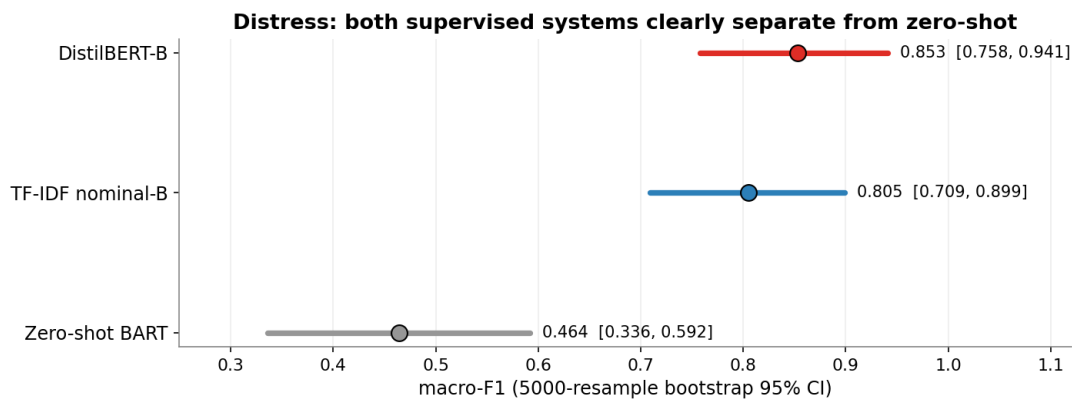


Figure 6: Distress: per-system macro-F1 with 95% bootstrap CIs (5000 resamples). Both supervised systems separate from zero-shot; the two supervised systems do not separate from each other. The DistilBERT-B estimate uses the five-seed majority vote.

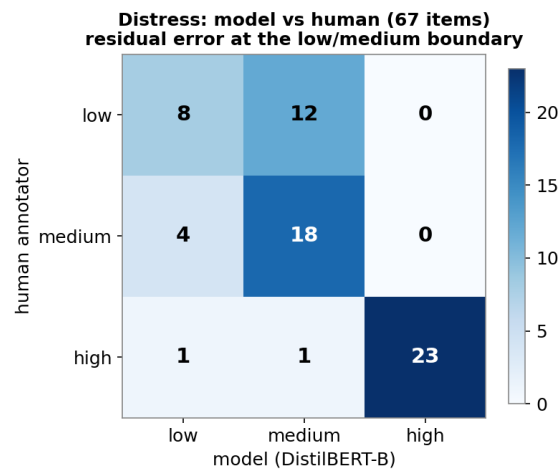


Figure 7: Distress: confusion between the best model (DistilBERT-B) and the blind human annotator on the 67-item test set. The residual error is concentrated at the low/medium boundary.

6 Error Analysis

Three recurring patterns emerged. (i) **Response — anger artifact:** anger has high agreement with the strict synthetic label but low agreement with the independent human reading; the model appears more closely aligned with the dual-judge annotation policy than with the independent annotator's reading (in oncology text, anger is often expressed as grief or protest that a human reads as sadness). (ii) **Response — anxiety over-prediction:** training on the consensus tier (C) inflates the predicted/support ratio for

anxiety; the cause is data composition, not the class weights. (iii) **Distress — low/medium boundary**: the dominant disagreement, both against the human and across models, is the systematic human-low → model-medium shift noted above. Across the trained systems, severe errors are nearly absent and high distress is detected well, so the residual difficulty is semantic ambiguity at an adjacent boundary rather than an extreme ordinal error.

7 Discussion

The two tasks responded differently to model family and synthetic-data tier. Response labels were highly accessible to a sparse lexical classifier, which matched the fine-tuned transformer; distress showed a larger descriptive advantage for the transformer and a much larger gap between supervised and zero-shot systems. This suggests distress benefits more from contextual fine-tuning than response classification does — yet the lexical model still reached 0.805 on distress, so this is a descriptive advantage, not evidence that distress *requires* a transformer. Practically, a TF-IDF model is a strong, cheap, and interpretable baseline that should not be skipped, and reporting bootstrap CIs is essential: several mean differences that look decisive are not statistically resolved at a realistic test-set size.

8 Limitations

Several limitations bound our claims. The corpus is synthetic, so its class distributions reflect the generation pipeline rather than real-world prevalence, and findings should be confirmed on real text. The human-reference evaluation used a single annotator and therefore measures agreement with one independent reading rather than with a multi-annotator gold standard. The test set has 67 items, which limits the power to separate close systems (as the bootstrap makes explicit). Finally, the low/medium distress distinction is inherently subjective and was the least reliable part of the label scheme. The resulting classifiers are research prototypes evaluated on synthetic text and are not validated for clinical triage or individual-level decision-making.

9 Conclusion

We defined a novel two-label oncology task and built a fully synthetic, attribute-controlled, label-leakage-controlled corpus for it, audited by two independent LLM judges into quality tiers. Across three model families, task-specific synthetic supervision was consistently useful, but its value depended on both the task and the model family: a sparse lexical model was highly competitive for response, while distress showed a larger (though not statistically resolved) advantage for contextual fine-tuning. Increasing the amount of consensus-labeled data did not uniformly improve performance. Human validation showed that the largest residual errors were concentrated at a semantically ambiguous boundary (low/medium distress) rather than at severe failures. Future work includes a multi-annotator gold standard, a more context-aware distress model, and validation on real (non-synthetic) text.

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